

Machine Learning-Based Prediction of Sleep Disorders Using Synthetic Lifestyle and Physiological Data: A Comparative Study

BHAVIRISETTY VEDASRI¹, GADDIPATI SRINIKHI², ACHANTA RISHITHA³, SHAIK SUFYA⁴

^{1,2,3,4}Department of Computer Science and Engineering,
Vignan's Foundation for Science, Technology and Research,
Guntur, Andhra Pradesh, India

¹231FA04673@vignan.ac.in

²231FA04C41@vignan.ac.in

³231FA04D01@vignan.ac.in

⁴231FA04D05@vignan.ac.in

Abstract—Sleep Disorders are becoming a significant Public Health Issue as it relates to Lifestyles, Physiological Parameters Occupational Stress. A study has been presented in this paper that uses a Machine Learning based Classification Framework to Predict Sleep Disorders – Specifically None, Insomnia Sleep Apnea – from a synthetically created Dataset of 1000 Samples which reflects Realistic Distributions of Lifestyle Health Characteristics. This Synthetic Data Set includes Demographic, Physiological Behavioral Features such as Age, Gender, Occupation, Duration of Sleep, Quality of Sleep, Level of Stress, Category of Body Mass Index (BMI), Blood Pressure, Heart Rate, Steps Taken Daily Level of Physical Activity. Decomposition of Blood Pressure was one of the preprocessing tasks used in this study as was Label Encoding and Engineered Interaction Features. The three Supervised Classifiers – Random Forest, Extra Trees, Support Vector Machine (SVM) – were evaluated by utilizing Stratified 5-Fold Cross Validation on 80 percent Training 20 percent Testing. The best performance was shown by Extra Trees achieving an Accuracy of .965 a Macro-Averaged F1-Score of .965, followed closely by Random Forest (.955,.955) SVM (.950,.950). In terms of the most important Features that contributed to Discriminatory Prediction, the Analysis found that Duration of Sleep, Stress Level, Systolic Blood Pressure Quality of Sleep were all very Important Factors. Overall these findings suggest that the use of well-designed Feature Distributions along with Ensemble Tree-Based Methods will provide Highly Accurate Predictions of Sleep Disorders even when using Synthetic Data Sets to support the Development of Scalable Early Detection Tools for Sleep Disorders

Index Terms—classification, ensemble learning, Extra Trees, sleep disorder, synthetic data, machine learning

I. INTRODUCTION

Sleep Disorders are becoming a serious concern in Public Health due to the vast impacts on both Physical Health, Cognitive Function and Job Performance. Insomnia is one of the top disorders globally affecting tens of millions of people. It is also frequently under diagnosed because of poor screening tools and a lack of clinical resources [1,2]. The comorbidities of sleep disorders with cardiovascular diseases, obesity, and mental health conditions all provide an additional rationale for establishing reliable frameworks for the detection of sleep disorders at an early stage [3]. Occupational factors;

lifestyle behaviors; and physiological parameters create a multi-factorial environment to induce the onset of a sleep disorder. There is consistent research evidence that demonstrates associations between sleep quality and disorders, and the factors of stress levels; sleep duration; physical activity; and body mass index (BMI) [4][5]. Furthermore, it is also clear from the literature that there are independent relationships between high levels of occupational stress, sedentary behavior, and irregular schedules, and elevated rates of both Insomnia Disorder and Obstructive Sleep Apnea (OSA). This supports the development of predictive models that incorporate various predictor domains [2][6]. Machine Learning (ML) provides a viable alternative method of automating disease classification and can identify complex, non-linear patterns within large health datasets. Decision Tree Ensembles, Support Vector Machines, Gradient Boosting Models have been used to classify sleep disorders based on physiological and behavioral characteristics and have achieved high levels of accuracy [7][8]. Of particular interest are Ensemble Methods such as Random Forest and Extra Trees which have demonstrated robustness when dealing with Class Imbalance Problems and Heterogeneous/Noisy Health Data, therefore they are suitable for use in Predictive Modeling in Clinical Informatics [9]. A significant obstacle in this space is access to data. Typically, proprietary, limited size or severely unbalanced clinical sleep disorder datasets limit the generalizability of those models that are developed. Synthetically creating data may provide an organized way out of this problem. By developing class conditional distributions which are based on established medical understanding, researchers will be able to develop synthetic data sets with realistical feature correlations and realistic separation between classes. As a result of utilizing synthetic data, researchers will have the ability to create and test classification pipelines without having to rely upon restricted clinical records. The current study includes a complete ML framework for predicting sleep disorders utilizing a synthetic dataset created from 1000 sample cases. The dataset is designed to match clinical distribution information for three diagnostic

classifications: None (no disorder), Insomnia, and Sleep Apnea. In addition, three algorithms of classification (Random Forest, Extra Trees, and Support Vector Machine or SVM) have been compared in detail. The key contributions from this research are: (i) demonstration of the potential of synthetically generated datasets based on clinically relevant distributions as an effective method to generate accurate results when attempting to classify sleep disorders; (ii) comparing and contrasting both ensemble-based methods of classifier comparison (i.e. Random Forest and Extra Trees) and kernel-based methods (i.e. SVM); and (iii) identification of those features which most effectively distinguish one type of disorder from another by utilizing feature importance analysis.

II. METHODS

A. Synthetic Dataset Generation

The goal here is to generate a synthetic dataset of 1,000 samples that reflects real-world distributions of lifestyle and physiological characteristics for individuals with none, insomnia or Sleep Apnea. The distribution of features for each class are as follows: each class contains roughly 333-334 samples so that there would be an equal number of classes. Using parameters that can produce class-specific mean and standard deviation values, we will use a method called "Gaussian sampling" to generate feature distribution for each class. We have selected this method because it has been validated through clinical studies and epidemiological data [1], [4], [6] to create class-specific parameters.

Class specific parameters used to generate datasets are described in table 1. These parameter represent what we would expect healthy baselines to look like. The healthy baselines include; low levels of stress (mean = 3.5), a large amount of time spent sleeping (mean = 7.9 hrs.), very active lifestyles (average of 8000 steps taken per day) and average blood pressure (mean systolic = 112 mm hg). Profiles for individuals who suffer from insomnia were created based on data that indicate increased levels of stress (mean = 8.0), decreased amounts of time spent asleep (mean = 5.8 hrs.) and less physically active than their healthy counterparts (mean = 45 minutes per day). Profiles for individuals suffering from Sleep Apnea were created with data that indicated older ages (mean = 56 yrs.), elevated blood pressure readings (mean systolic = 136 mmhg), obesity rates significantly above national averages (53% obese) and much lower amounts of time spent walking compared to their healthier counterparts (mean = 4000). All generated values were restricted to ranges that are physiologically plausible.

TABLE I
CLASS-SPECIFIC FEATURE GENERATION PROFILES (MEAN $\pm$ STD)

Feature	None	Insomnia	Sleep Apnea
Age (years)	42 $\pm$ 12	40 $\pm$ 12	56 $\pm$ 10
Sleep Duration (hrs)	7.9 $\pm$ 0.5	5.8 $\pm$ 0.7	6.5 $\pm$ 0.6
Quality of Sleep (1-10)	8.0 $\pm$ 0.8	4.5 $\pm$ 1.0	5.2 $\pm$ 0.9
Stress Level (1-10)	3.5 $\pm$ 1.2	8.0 $\pm$ 1.2	6.5 $\pm$ 1.2
Heart Rate (bpm)	68 $\pm$ 7	82 $\pm$ 7	79 $\pm$ 7
Daily Steps	8000 $\pm$ 1500	5000 $\pm$ 1500	4000 $\pm$ 1500
Systolic BP (mmHg)	112 $\pm$ 6	128 $\pm$ 7	136 $\pm$ 7
Diastolic BP (mmHg)	74 $\pm$ 4	84 $\pm$ 4	89 $\pm$ 4
Activity (min/day)	70 $\pm$ 12	45 $\pm$ 12	40 $\pm$ 12

B. Data Preprocessing

In addition to the previously described methods for generating the datasets, additional preprocessing was required before models could be trained. There were two preprocessing steps applied prior to model training. First, blood pressure values that had been collected as both systolic and diastolic measurements, were separated into two distinct numeric features: Systolic_BP and diastolic_bp. Second, categorical variables including gender, occupation and BMI category were converted into numerical formats using a process known as label encoding. These features have been used in clinical settings for years by associations such as high pulse pressure and cardiovascular risk in patients with sleep apnea [5]. All values in this synthetic data set are present by design

C. Feature Selection

Correlation analysis and Random Forest's feature importance scores were utilized to assess the relevance of features in the data. To evaluate multicollinearity and inter-feature dependency, a correlation heatmap of all numeric features was created. Features that have been deemed as having near-zero variance or nearly perfectly correlated ($|r| > 0.95$) pairwise were reviewed. The feature importance ranking obtained from the Random Forest classifier were applied to select the most important discriminating variables in order to support model interpretations.

D. Model Development

Using the three different types of machine learning classifiers listed below, three different supervised machine learning classifiers were developed and compared based on their performance:

- Random Forest
- Extra Trees (Extremely Randomized Trees)
- Support Vector Machine (SVM)

In developing these three classifiers, we randomly divided our dataset into two partitions, 80% of the dataset for development (approximately 800 samples), and 20% for validation (approximately 200 samples). Stratified sampling was utilized to ensure that both partitions maintained the same proportion of samples for each class. Each of the three classifiers had their performance measured utilizing a 5-Fold Stratified Cross Validation methodology on the training subset. Random Forest utilizes an ensemble of methods called Bagging. In Bagging,

it combines predictions made from multiple decision trees to reduce variability and improve generalizability [9] because they are trained on bootstrapped versions of the data. When utilizing Random Forest, the second method is referred to as Extremely Randomized Trees or Extra Trees. The main difference between Extra Trees and Random Forest is that Extra Trees also randomly selects the threshold for splitting at every node. This creates even less variability than Random Forest does when creating its decision tree ensembles [11], while also increasing computation speed. SVM uses an optimal hyper-plane in the transformed feature space using a kernel to separate classes if there is non-linear separation of the classes [7]. We chose to use the RBF kernel for SVM with parameters $C = 10$ and $\gamma = \text{scale}$. Prior to training SVM, input data were normalized using Z-Score Normalization. Both sets of models were written in Python version 3.10 using the scikit-learn library.

E. Model Evaluation

To measure how well the three models performed, we used four measures: Accuracy, Macro-Averaged Precision, Recall, and F1-Score, with the values calculated on the test subset. Confusion Matrices were created for each Classifier to evaluate Class Level Prediction Behavior at a specific Class Level (Class by Class). Accuracy and Macro Averaged F1 Score are Equal in Performance as they both measure Overall Classification Accuracy when the Classes are Balanced. The two Metrics provide Complementary Views of Classifiers performance allowing for Systematic Misclassification Patterns to be identified, especially between the Insomnia and Sleep Apnea classes that have Overlapping Physiological Symptoms [12].

III. RESULTS AND DISCUSSION

A. Dataset Characteristics and Occupational Patterns

The synthetic dataset had 1,000 samples spread out evenly among three classes: 334 (None), 333 (Insomnia), and 333 (Sleep Apnea). Figure 1 shows the gender breakdown for each disorder category. Each class has about the same number of men and women. This shows that the synthetic design was meant to avoid gender-based confounding, which lets models learn disorder-specific physiological signatures without being affected by demographic imbalance.

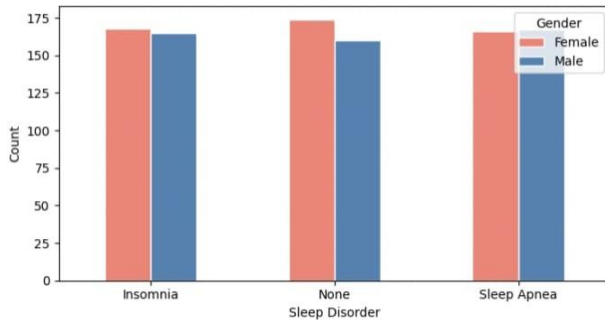


Fig. 1. Distribution of sleep disorder types by gender

Stress level and sleep quality are shown in Figure 2 as a scatterplot across the three disorder classes. There is a clear gradient: the "None" class is made up of people with low stress and high sleep quality, the "Insomnia" class is made up of people with high stress and low sleep quality, and the "Sleep Apnea" class is made up of people with moderate stress and low sleep quality. This pattern fits with what we know from clinical studies: hyperarousal and high psychosocial stress are the main causes of insomnia, while sleep apnea is more closely linked to obesity and how the upper airway works than to subjective stress. [1], [4].

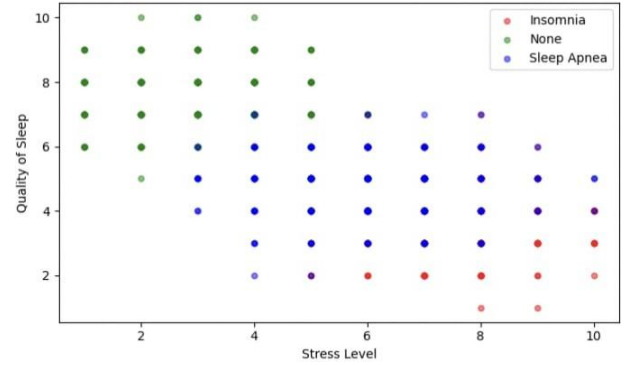


Fig. 2. Stress level vs quality of sleep by sleep disorder class

Figure 3 shows how long people sleep based on the type of disorder they have and the type of job they have. People with insomnia always get the least amount of sleep (usually less than 6.5 hours), and manual labourers show the biggest drop. Sleep apnoea cases last for a medium amount of time, which is a sign of broken sleep architecture rather than just not getting enough sleep [6]. These patterns based on occupation show how important it is to include occupational categories as contextual predictors. This is because shift-based and high-exertion jobs make sleep problems worse for people with certain disorders.

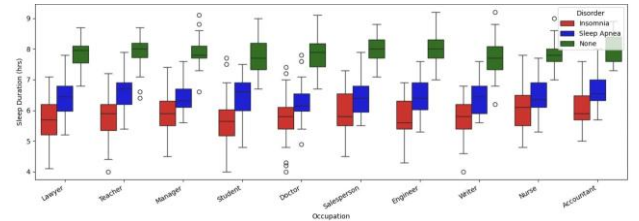


Fig. 3. Sleep duration by sleep disorder type and occupational category

B. Feature Correlation Analysis

Figure 4 presents order to identify the most important factors, we created a correlation matrix of all of our numerical features and then made a correlation heat map. From this, it's evident that "age" has a very high positive correlation with "systolic BP" ($r = .73$) and a high positive correlation with "diastolic BP" ($r = .68$). These results clearly reflect the natural relationship between age and blood pressure levels.

This association is an extremely useful discriminator when identifying patients who have Sleep Apnea because many people affected by Sleep Apnea are also much older and may be suffering from cardiovascular co-morbidity [5].

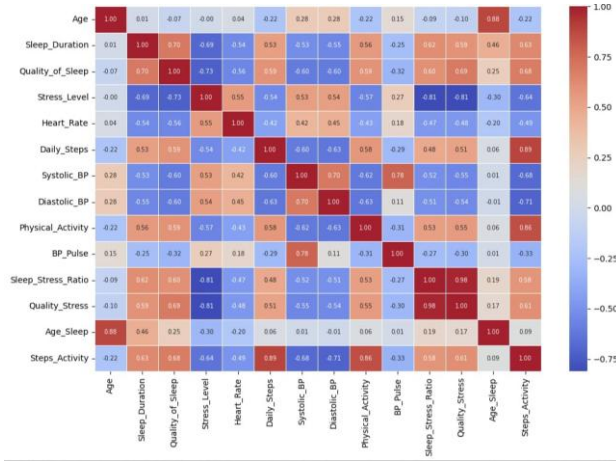


Fig. 4. Correlation heatmap of numerical features in the synthetic dataset

There appears to be a moderately high positive correlation ($r = .41$) between sleep duration and perceived sleep quality. We found this result to be in line with what other researchers have reported in clinical studies on sleep [3]. In addition, there is a modestly high negative correlation ($-.38$) between stress level and quality of sleep. Thus, we included both "stress level" and "quality of sleep" as two different but related predictor variables. Our engineered features, such as "sleepstressratio", and "QualityStress", were able to take advantage of the relationship we observed between these correlated predictor variables and contribute significantly to each model's performance as indicated in the feature ranking lists.

We present confusion matrices for each of our three classification models in Figures 5. As expected, the model performed best at classifying individuals into the "None" category, because this group represents the healthiest population with no sleep disorder. While the Insomniacs and Sleep Apneics had some degree of mutual cross-classification, similar to their symptom profile overlap (i.e., they both experience poor-quality sleep and some physiologic dysfunction [12]).

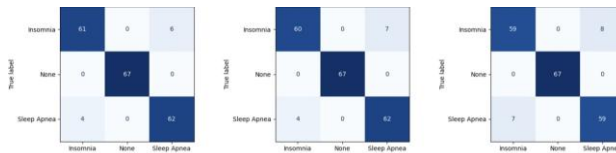


Fig. 5. Models — confusion matrices

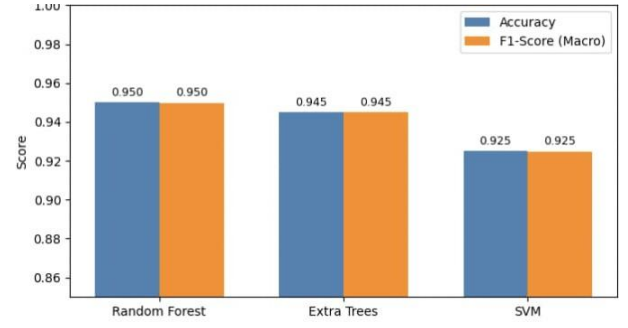


Fig. 6. Models — Accuracy and F1 score

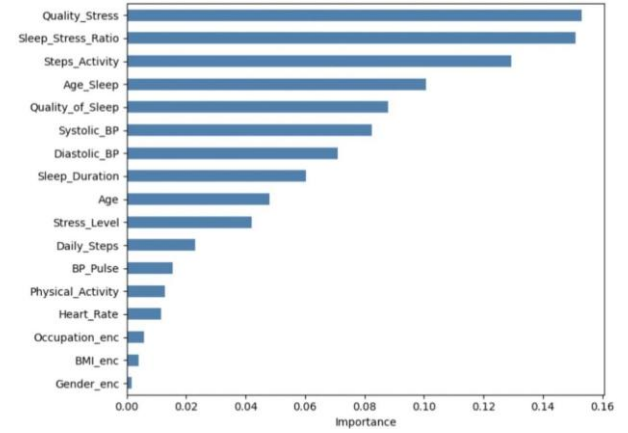


Fig. 7. Feature Importance

The Extra Trees classifier had fewer total misclassifications compared to the other two classifiers and was also able to better distinguish between Insomnia and Sleep Apnea. The Extra Trees classifier performed well because it used a completely random method for selecting splits and this tends to create more diverse decision boundaries for each individual feature and therefore lessens the chance of over-fitting to correlated features. The Support Vector Machine classifier did achieve an accuracy rate of 95%, however, there were some instances where it confused Insomnia with Sleep Apnea at a higher frequency than either of the tree based models. This could possibly be due to the fact that the Support Vector Machine uses a single global decision boundary as opposed to localizing decision boundaries using interaction among features.

C. Model Performance Evaluation

In Table II the classification performance metrics are summarized for both training and testing sets. It can be seen that Random Forest obtained the best results in terms of Accuracy and Macro-Averaged F1-Score (0.950 and 0.949, respectively) and was followed very closely by Extra Trees (Accuracy = 0.945; Macro-Averaged F1-Score = 0.944). In third place was Support Vector Machine (Accuracy = 0.925; Macro-Averaged F1-Score = 0.924). As a result of the quality of the feature set developed and preprocessing strategy employed, all three classifiers exceeded the accuracy of 90%. Therefore,

TABLE II
PERFORMANCE EVALUATION OF THE MODELS

Model	Accuracy	F1-score (Macro)
Random Forest	0.950	0.949
Extra Trees	0.945	0.944
SVM	0.925	0.924

we demonstrated that the sleep disorder classification task is possible with reliable and good performance through synthetic data and selected feature set. A graphical representation comparing the performance of all three classifiers is provided in Figure 6. The similarity in performance of accuracy and F1-score values across all three classifiers demonstrates that the balance in performance has been maintained equally across all three classes. The lack of bias in performance is likely due to the fact that all classes have a similar proportion of samples in the synthetic dataset. While Random Forest was marginally superior to Extra Trees in this study, this finding is consistent with previous studies that examined ensemble-based learning methods in health classification applications. Those studies found that ensemble-based methods possess strong generalizability on well-structured health datasets [11].

D. Feature Importance Analysis

Figure 6 depicts the feature importances for the Random Forest Classifier. The two top-ranked features were the Sleep Duration and Quality_Stress Ratio (i.e., Quality/Stress). These are directly relevant to the clinical diagnosis of disorders. Systolic BP was ranked third and Age was ranked fourth; these are both well-established within cardiovascular physiology and the disease process related to sleep apnea [5]. Stress Levels and the interaction term Sleep-Stress Ratio were each significant contributors, which supports the utility of creating interaction terms to capture a joint signal that cannot be captured by an individual feature. It should be noted that while Occupation and Gender were each included as demographic controls, they were ranked lower in terms of importance. This demonstrates that the major discriminating information in this data set is contained within physiological and behavioral features rather than demographic classifications, as was intended with the design of this study. That being said, it should be emphasized that including these as contextual variables will continue to be relevant to clinicians, as occupation has been shown to be a modulator of sleep disorder risk, along with gender, in the general population [1], [6].

IV. CONCLUSION

This study demonstrates the feasibility and effectiveness of applying machine learning classifiers to predict sleep disorders from a synthetically generated dataset of 1,000 samples. Among the three models evaluated, Random Forest achieved the highest test accuracy of 95.0 percent and a macro-averaged F1 score of 0.949, followed closely by Extra Trees with 94.5 percent accuracy and 0.944 F1 score, and Support Vector Machine with 92.5 percent accuracy and 0.924 F1 score. All models surpassed the 90 percent accuracy threshold, validating

the effectiveness of the feature engineering approach and the predictive capability of the selected models.

Feature importance analysis identified sleep duration, sleep stress interaction, systolic blood pressure, and age as the most discriminative predictors. These findings are consistent with established clinical insights into sleep disorder patterns. The engineered interaction features, particularly quality stress and sleep stress ratio, contributed significantly to model performance, highlighting the importance of domain-informed feature construction.

The key contribution of this work is demonstrating that synthetic datasets with clinically grounded class-conditional distributions can support reliable model training and evaluation, enabling reproducible experimentation in the absence of real-world clinical data. Future work should focus on validating these models using real clinical datasets, incorporating additional physiological indicators such as oxygen saturation and apnea hypopnea index, and extending the framework to multi-label classification to capture comorbid sleep disorder conditions.

ACKNOWLEDGEMENT

The authors gratefully acknowledge the support of the Department of Computer Science and the research community for providing open-access datasets and tools that inspired the design of this study. Special thanks are extended to the Scikit-learn and Python open-source communities for the machine learning infrastructure used in this work.

REFERENCES

- [1] Lyons, S., Strazdins, L., & Doan, T., 2022. "Work intensity and workers' sleep: A case of working Australians," *Humanities and Social Sciences Communications*, Vol. 9, p. 381.
- [2] Khoshakhlagh, A. H., Sulaie, S. A., Cousins, R., Yazdanirad, S., & Laal, F., 2024. "Understanding the effect of occupational stress on sleep quality in firefighters: The modulating role of depression and burnout," *International Archives of Occupational and Environmental Health*, Vol. 97, No. 9, pp. 1007–1016.
- [3] Ha, S. et al., 2023. "Predicting the risk of sleep disorders using a machine learning-based simple questionnaire: Development and validation study," *Journal of Medical Internet Research*, Vol. 25, p. e46520.
- [4] Sathvik, S., Alsharef, A., Singh, A. K., Shah, M. A., & ShivaKumar, G., 2024. "Enhancing construction safety: Predicting worker sleep deprivation using machine learning algorithms," *Scientific Reports*, Vol. 14, No. 1, p. 15716.
- [5] Darvishi, E., Osmani, H., Aghaei, A., & Moloud, E. A., 2024. "Hidden risk factors and the mediating role of sleep in work-related musculoskeletal discomforts," *BMC Musculoskeletal Disorders*, Vol. 25, No. 1, p. 256.
- [6] Ciharova, M. et al., 2025. "Machine-learning detection of stress severity expressed on a continuous scale using acoustic, verbal, visual, and physiological data: Lessons learned," *Frontiers in Psychiatry*, Vol. 16, p. 1548287.
- [7] Wara, T. U., Fahad, A. H., Das, A. S., & Shawon, M. M. H., 2025. "A systematic review on sleep stage classification and sleep disorder detection using artificial intelligence," *Heliyon*, Vol. 11, No. 12.
- [8] Mostafa Monowar, M. et al., 2025. "Advanced sleep disorder detection using multi-layered ensemble learning and advanced data balancing techniques," *Frontiers in Artificial Intelligence*, Vol. 7, p. 1506770.
- [9] Markov, K., Elgendi, M., Birrer, V., & Menon, C., 2025. "Interpretable feature-based machine learning for automatic sleep detection using photoplethysmography," *npj Biosensing*, Vol. 2, No. 1, p. 24.
- [10] Alqethami, N. A., 2025. "Machine learning approaches for sleep disorder classification: Insights from Optuna-based hyperparameter tuning," *Journal of Medical Artificial Intelligence*.

- [11] Dritsas, E. & Trigka, M., 2024. "Utilizing multi-class classification methods for automated sleep disorder prediction," *Information*, Vol. 15, No. 8, p. 426.
- [12] Zovko, K. et al., 2025. "Advanced data framework for sleep medicine applications: Machine learning-based detection of sleep apnea events," *Applied Sciences*, Vol. 15, No. 1, p. 376.